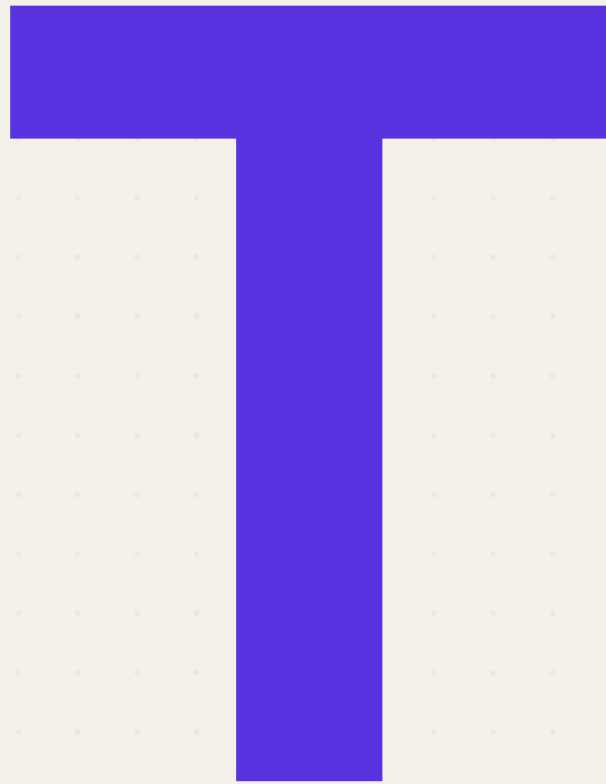




LV.0 → LV.1 · AI ENGINEER

The 6 T's that move you from **Level 0** → **1** AI engineer.

Transformers, Tokenization, Temperature, Tool Calling and more.
Plain-English definitions, one swipe each, plus the tool and the
mind to follow.

TRANSFORMERS • ARCHITECTURE

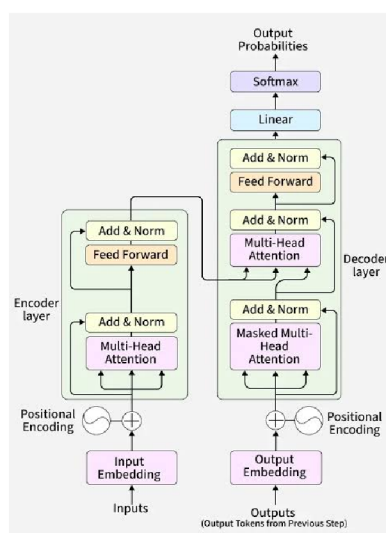
Transformers

attention is all you need • 2017

The Transformer is the architecture behind every modern LLM, using **self-attention to weigh how tokens relate**, processing sequences in parallel. GPT, Claude, Gemini, and Llama are all built on it.

USED IN

- LLMs
- Vision (ViT)
- Multimodal AI



ENCODER • DECODER • ATTENTION

Tokenization

text · → · tokens

Models don't read words. They read tokens. Tokenization **breaks text into the units a model processes**, driving context length, cost, and speed. "ChatGPT is amazing!" becomes ["Chat", "GPT", " is", " amazing", "!"].

USED IN

- BPE / tiktoken
- Context windows
- Pricing

"ChatGPT is amazing!"



Chat

GPT

is

amazing

!

WORDS → SUBWORD TOKENS

TEMPERATURE • GENERATION

Temperature

focused ← → creative

Temperature is the dial for **randomness in a model's output**. Low (0–0.3) is deterministic and factual, great for code. High (1.0+) is diverse and creative, great for brainstorming. Same prompt, very different answers.

USED IN

- Sampling
- Prompt engineering
- API params



0.0–0.3

deterministic
factual • code

0.5–0.8

balanced
general use

1.0+

creative
brainstorm • story

RANDOMNESS DIAL

TF-IDF

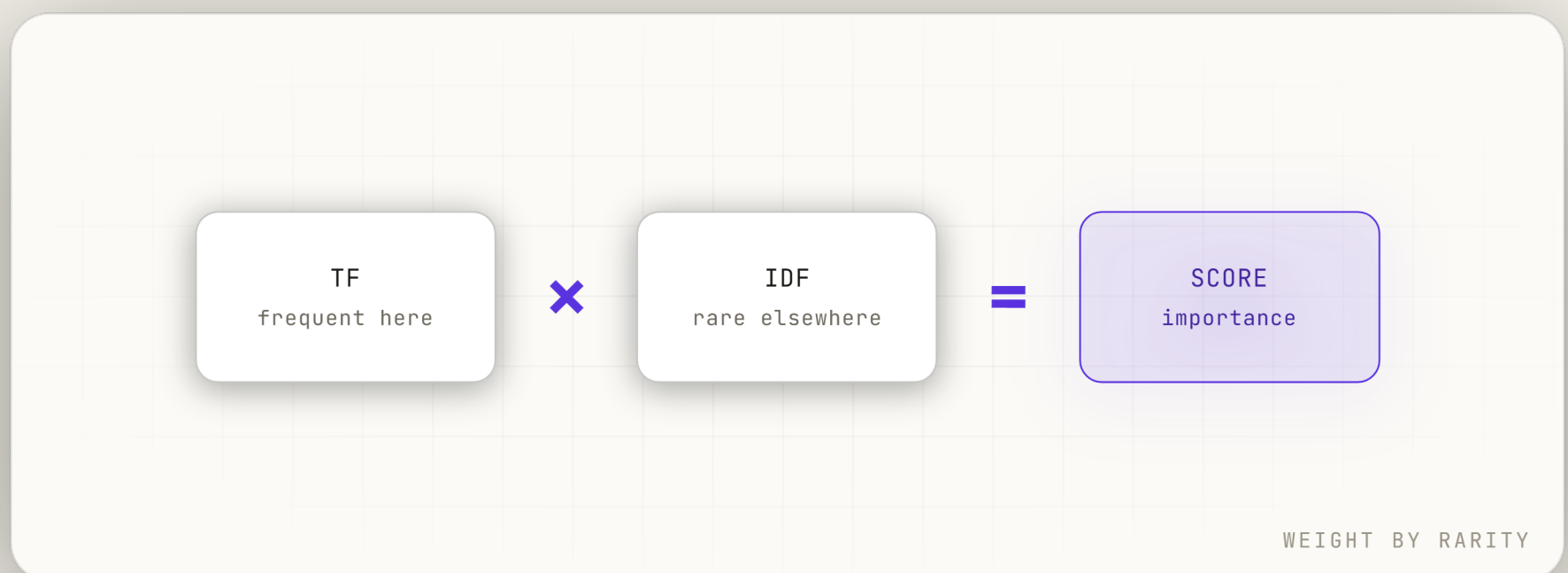
term freq × inverse doc freq

TF-IDF scores how important a word is to a document by **rewarding terms frequent here but rare elsewhere**.

"Election" scores high in a politics piece, low in sports, the classic, interpretable backbone of keyword search.

USED IN

- Search engines
- Keyword extraction
- Hybrid search



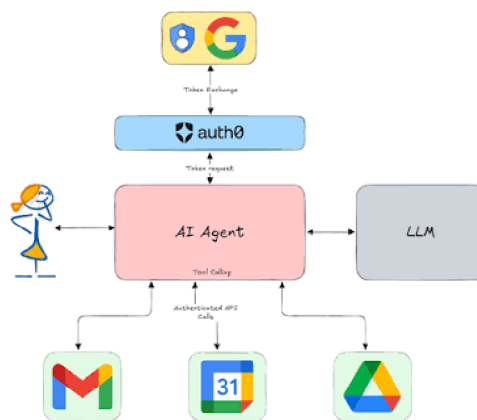
Tool Calling

model → real actions

Tool calling lets a model **invoke real functions and APIs**, search, email, calendars, databases, instead of only generating text. It's what turns a chatbot into an agent that gets things done.

USED IN

- AI agents
- Function calling
- MCP



AGENT → AUTHENTICATED APIS

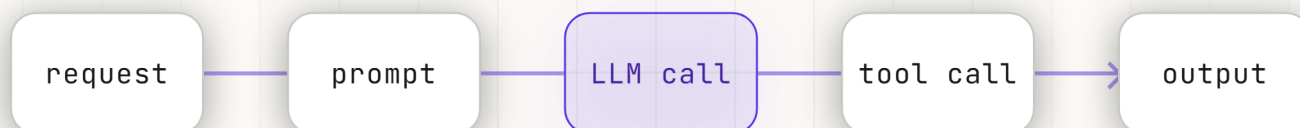
Tracing

a flight recorder • for AI

Tracing records **every step an AI takes**, prompts, model calls, tool use, latency, tokens, and errors, so you can debug, cut costs, and reduce hallucinations in multi-step agents instead of guessing from the final answer.

USED IN

- LLMOps
- LangSmith / Langfuse
- Agent debugging



🕒 latency • 📄 tokens • ⚠ errors logged at every hop

FULL EXECUTION PATH

↓ UP NEXT · TOOL

Tool next.

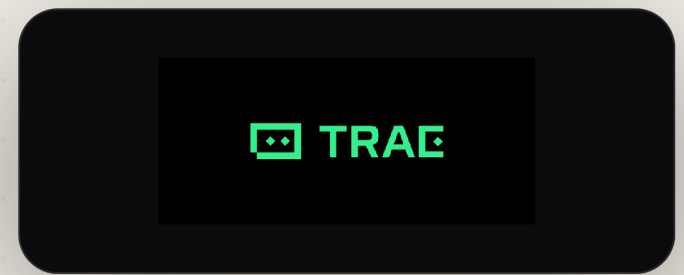
Beyond the **6 T-words**, an AI-native IDE
where agents write, refactor, and ship code
alongside you.



AI-NATIVE CODING IDE

Trae

by ByteDance • agentic IDE



Trae is an AI-native IDE where **agents understand your project and execute coding tasks** in natural language, write, edit, refactor, and generate, powered by Claude, DeepSeek, and GPT-class models.

WHAT IT IS

An AI-native IDE with built-in agents.

IN AI STACKS

Project-aware code gen & refactoring.

WHY DEVS USE IT

Beyond autocomplete, full-task coding.

PAIRS WITH

Claude, DeepSeek, GPT-class models.

BRAIN-RESPONSE FOUNDATION MODEL

TRIBE v2

Meta AI • fMRI-trained brain encoder

Meta AI's Trimodal Brain Encoder, a foundation model trained on **1,000+ hours of human fMRI data** to predict how the brain responds to images, sound, and language, a step toward AI that perceives more like we do.

WHAT IT IS

A foundation model of human brain responses.

BUILT ON

1,000+ hrs of fMRI from many subjects.

WHY IT MATTERS

Predicts neural activity from text, image, sound.

PAIRS WITH

Neuroscience, multimodal AI research.

↓ UP NEXT · PERSON TO FOLLOW

Spotlight next.

The **high-signal, low-noise** educator worth a follow to keep your T-stack current.



HIGH-SIGNAL, LOW-NOISE

Tina Huang

YouTube • @TinaHuang1 • ex-Meta



Former Meta data scientist turned educator, teaching **AI, data science, and self-learning strategies** to 1M+ followers, practical, career-focused content for engineers leveling up.

WHAT THEY POST

AI, data science & tech-career content.

FIND THEM

YouTube @TinaHuang1, LinkedIn.

WHY FOLLOW

Practical, build-and-learn strategies.

BEST FOR

Self-teaching AI & data science.

HIGH-SIGNAL, LOW-NOISE

Tomas Mikolov

creator of Word2Vec • ex-Google, FAIR



The researcher who gave NLP its first practical word vectors , **Word2Vec** and **FastText**, making meaning computable and inspiring every embedding-based system since.

KNOWN FOR

Word2Vec, FastText, neural LM.

FIND THEM

Google Scholar, BottleCap AI.

WHY FOLLOW

Foundational ideas behind embeddings.

BEST FOR

How language became math.

HIGH-SIGNAL, LOW-NOISE

Tech with Tim

Tim Ruscica • YouTube educator



A leading **project-first programming & AI engineering channel**, practical tutorials that take you from concept to a shipped AI app.

KNOWN FOR

Python, ML, LLM apps, AI agents.

FIND THEM

YouTube @TechWithTim, LinkedIn.

WHY FOLLOW

Build-and-ship style tutorials.

BEST FOR

Going from theory to product fast.

HIGH-SIGNAL, LOW-NOISE

Thomas Kurian

CEO • Google Cloud



The executive who turned **Google Cloud** into a serious enterprise AI platform, scaling Vertex AI, Gemini-in-Workspace, and AI infra for the Fortune 500.

KNOWN FOR

Google Cloud, Vertex AI, Gemini.

FIND THEM

X @ThomasOrTK, LinkedIn.

WHY FOLLOW

How big enterprises actually buy AI.

BEST FOR

AI strategy at enterprise scale.

HIGH-SIGNAL, LOW-NOISE

Tom Brown

co-founder • Anthropic



Lead author of the [GPT-3 paper](#) and Anthropic co-founder, one of the engineers who proved that scale alone could turn LLMs into few-shot learners.

KNOWN FOR

GPT-3, scaling, Anthropic, Claude.

FIND THEM

X @nottombrown, LinkedIn.

WHY FOLLOW

Built the model that started the boom.

BEST FOR

Scale, training, frontier LLMs.

HIGH-SIGNAL, LOW-NOISE

Tiffany Janzen

founder • TiffinTech



Software dev turned creator running **TiffinTech and Talk Tech with Tiff**, making AI, tooling, and tech careers accessible across socials.

KNOWN FOR

TiffinTech, dev education, podcasts.

FIND THEM

IG @tiffintech, YouTube.

WHY FOLLOW

Approachable take on AI & careers.

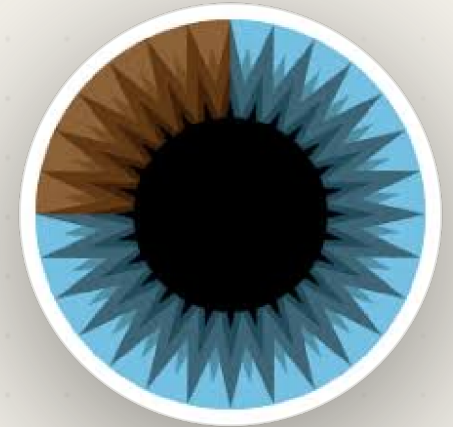
BEST FOR

Career-focused AI content.

HIGH-SIGNAL, LOW-NOISE

3Blue1Brown

Grant Sanderson • math made visual



The channel that **animates the math behind AI**, neural nets, backprop, transformers, gradient descent, so intuition arrives before the equations.

KNOWN FOR

Neural net visualizations, math.

FIND THEM

YouTube @3Blue1Brown.

WHY FOLLOW

See AI math instead of memorize it.

BEST FOR

Deep learning intuition.

HIGH-SIGNAL, LOW-NOISE

Tech with Tim

YouTube • AI engineering channel



Tim Ruscica's second feature here, the **channel itself**, with hands-on series on LLM apps, AI agents, and full-stack ML projects.

KNOWN FOR

LLM apps, AI agents, ML projects.

FIND THEM

YouTube @TechWithTim.

WHY FOLLOW

Long-form, follow-along builds.

BEST FOR

Building real AI apps end-to-end.

The T stack, in one card.

Save it for later. Drop a comment if there's a T-term you'd add.

01 Transformersarchitecture	02 Tokenizationnlp
03 Temperaturegeneration	04 TF-IDFretrieval
05 Tool Callingagents	06 Tracingobservability
+ Traetool	+ TRIBEv2tool
@ Tina Huangpeople	@ Tomas Mikolovpeople
@ Tech with Timpeople	@ Thomas Kurianpeople
@ Tom Brownpeople	@ Tiffany Janzenpeople
► 3Blue1Brownchannel	► Tech with Tim YTchannel



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

U Next